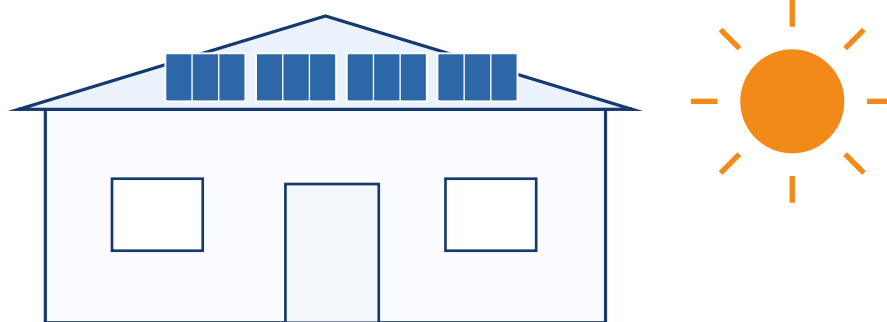


SOLAR ENERGY ASSESSMENT

Energy Assessment Report

A clear preliminary view of your recommended solar and battery system, expected savings and next steps.



Illustrative residential solar + battery concept

PROPERTY TYPE

Home

ASSESSMENT DATE

19 August 2026

PRIMARY OBJECTIVE

Reduce Diesel Use

PROJECT LOCATION

Benue, Nigeria

Recommendation snapshot

RECOMMENDED PV

3.3 kWp

BATTERY

14.4 kWh

ANNUAL SAVINGS

NGN 923,343

PAYBACK

7.1 years

EXECUTIVE SUMMARY

SolarVy assessed this **home in Benue, Nigeria** with the primary objective of reducing diesel use. The preliminary model recommends a **3.3 kWp solar PV system**, **14.4 kWh battery storage** and a **4.8 kW hybrid inverter**. Based on the assessment outputs, the system is estimated to generate **4,823.1 kWh of solar energy per year**, with net annual savings of approximately **NGN 923,343** and a simple payback of about **7.1 years**.

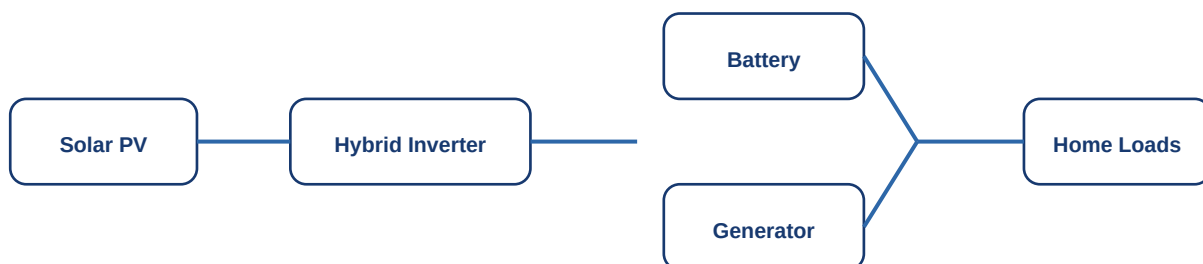
ENERGY PROFILE

Your energy profile & system logic

ENERGY PROFILE	ASSESSMENT VALUE
Estimated average monthly energy use	401.9 kWh/month
Estimated annual demand	4,823.1 kWh/year
Hybrid inverter rating	4.8 kW
Primary objective	Reduce Diesel Use
Assessment location	Benue, Nigeria

Data quality note: the previous report populated several energy-profile fields incorrectly. This improved version uses only consistent assessment values and clearly derived figures. Diesel displacement, energy independence and CO2 reduction are not shown as quantified results because the supplied report does not contain reliable values for them.

Illustrative system architecture



HOW THE SYSTEM WORKS

Solar PV supplies daytime household loads and charges the battery when surplus energy is available. The hybrid inverter manages power flow between solar, battery and the home. Battery storage supports the home during periods when solar generation is low or grid supply is unavailable. Generator support, where retained, should be treated as an auxiliary backup source rather than the primary daily energy source.

SYSTEM DESIGN

Recommended system & financial analysis

COMPONENT	RECOMMENDATION
Solar PV capacity	3.3 kWp
Battery storage	14.4 kWh
Hybrid inverter	4.8 kW
Annual PV generation	4,823.1 kWh
Usable solar energy	4,340.8 kWh

Financial summary

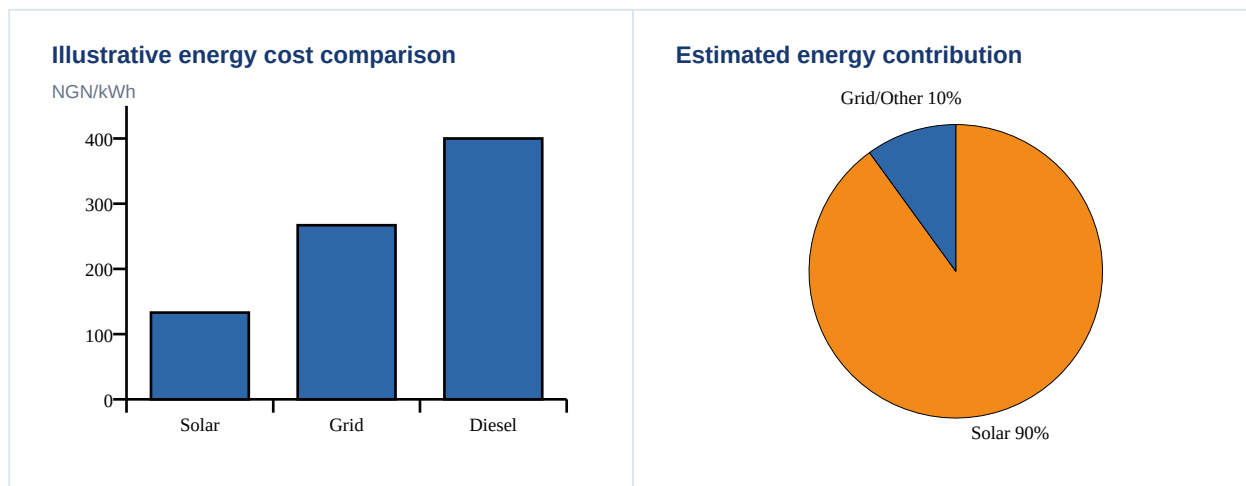
FINANCIAL METRIC	ESTIMATE
Total estimated system cost	NGN 6,538,560
Gross annual savings	NGN 1,054,114
Annual O&M; allowance	NGN 130,771
Net annual savings	NGN 923,343
Simple payback	7.1 years

FINANCIAL INTERPRETATION

The assessment indicates that the largest financial benefit comes from replacing more expensive conventional energy with solar generation. The simple payback period is an indicative planning metric and should be reviewed again once actual installer pricing, equipment warranties, operating costs and fuel usage are confirmed.

VALUE & NEXT STEPS

Energy economics and recommended next steps



SolarVy recommendation

A hybrid solar + battery system remains the appropriate preliminary direction for this assessment. The recommended battery capacity is materially larger than the PV array's average daily generation, so final engineering should verify the household's outage duration, critical loads and desired backup autonomy before procurement. This is particularly important because the stated objective is to reduce diesel use.

NEXT STEP	WHY IT MATTERS
1. Confirm outage and backup requirement	Verify typical hours without grid electricity and identify critical household loads.
2. Complete technical site review	Check roof area, orientation, cable routes, distribution board and protection requirements.
3. Obtain comparable installer quotations	Ask installers to quote against the same preliminary SolarVy sizing basis.
4. Review financial assumptions	Confirm actual equipment price, diesel cost, tariff and maintenance assumptions before committing.
5. Finalise engineering design	Installer or qualified engineer confirms final equipment selection and system protection.

Preliminary assessment disclaimer: This report is an indicative planning assessment generated from the supplied inputs and simplified assumptions. Final system sizing, equipment selection, electrical design, installation cost, energy production and financial performance should be validated by appropriately qualified professionals before procurement or installation.